



Effects of a professional development program on classroom practices and outcomes for Latino dual language learners

Virginia Buysse^{*,1}, Dina C. Castro¹, Ellen Peisner-Feinberg¹

FPG Child Development Institute, University of North Carolina at Chapel Hill, CB #8180, Chapel Hill, NC 27599-8180, United States

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ABSTRACT

A randomized, controlled study was conducted to assess the effects of the Nuestros Niños professional development program on classroom practices and child outcomes related to language development and early literacy skills in both English and Spanish. Fifty-five teachers and 193 Latino dual language learners (DLLs) enrolled in the North Carolina More at Four Pre-Kindergarten Program participated in the study. The content of the professional development program consisted of research-based instructional practices designed to complement the core curriculum and scaffold learning for DLLs, and the format included professional development institutes, individualized consultation, and community of practice meetings. The results showed that the intervention led to measurable improvements in both the overall quality of teachers' language and literacy practices as well as those specific to working with Latino DLLs, and greater gains in children's phonological awareness skills in their primary language.

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A large and rapidly growing number of children in the U.S. come from homes in which English is not the primary language. Latinos are now the largest minority group in the country, estimated to be 14.4% of the population, with an even larger percentage of Latinos among children below the age of five (Hernandez et al., 2007; U.S. Census Bureau, 2006). The rapid growth among Latinos over the past decade has raised questions about whether the existing education system can adequately address the needs and priorities of this population. On a variety of indicators such as end-of-grade assessments and high school graduation rates, Latino dual language learners (DLLs) are not faring as well as other students in U.S. schools. For example, one study involving 41 states found that only 18.7% of Latino students in a given school year scored above state-established norms in English reading comprehension (Kindler, 2002). Not surprisingly, English proficiency has been linked to school performance among Latino students. In a report published by the National Literacy Panel on Language-Minority Children and Youth, August and Shanahan (2006) concluded that the lower the level of English proficiency among Latino students, the lower their probability was of completing high school. In recent years, the expansion of early education programs to reach a wide range of young learners has emerged as a promising approach for promoting school readiness and ensuring future academic success for all children.

Nationally, Latinos are less likely than children from any other racial or ethnic group to be enrolled in an early education program (U.S. Department of Education, 2000). Compared with 73% of White and 76% of African-American first graders, only 57% of Latino first-graders had participated in center-based early childhood programs prior to kindergarten. One explanation for these lower participation rates comes from a national survey of state administrators of early childhood programs which

* Corresponding author. Tel.: +1 919 966 7171.

E-mail address: Virginia.Buysse@unc.edu (V. Buysse).

¹ Authors contributed equally to the development of this manuscript and are listed in alphabetical order.

reported that language difficulties and immigration status may serve as significant barriers for Latino families in terms of their full access and utilization of early childhood services (Buysse, Castro, West, & Skinner, 2005). A report issued by the National Council of La Raza noted that Latino children represented a large and growing proportion of the total enrollment in Head Start classrooms, accounting for more than one-third of all eligible children (Calderon, 2005). This finding suggests that the trend for Latino children and families to participate less often than other groups in early education programs may be changing, at least among children enrolled in Head Start.

However, even among those who have attended early childhood programs, Latino children lag behind their peers when they enter kindergarten and the gap in academic achievement appears to widen as children grow older (Reardon & Galindo, 2006). Among 3–5-year-olds not yet enrolled in kindergarten, White and African-American children were more likely than Latinos to recognize most letters of the alphabet, participate in storybook activities, count up to 20, and write or draw rather than scribble (U.S. Department of Education, 2000). Among a national sample of kindergartners participating in the Early Childhood Longitudinal Study (ECLS-K), Latino children had the lowest mean scores on reading proficiency compared to African-American, Asian, and White children; moreover, 50% of Latino children were not proficient in letter recognition, as compared to 43% of African-American, 21% of Asian, and 29% of White children (West et al., 2000).

More recent research suggests that Latino pre-kindergartners might benefit more than other groups from high-quality early childhood education (Gormley & Gayer, 2005). An evaluation of the Tulsa, Oklahoma Pre-K for All program showed greater gains for Latino pre-kindergartners than for White, Black, and Native American groups on key early literacy skills such as print awareness and letter identification (Barnett, Lamy, & Jung, 2005). The authors noted that these results were derived from children's enrollment in high-quality early education programs; the pre-kindergarten teachers were required to have at least a bachelor's degree and to implement specific instructional practices. As a result, findings from this study might not generalize to early education programs with less qualified teachers and lower overall program quality.

Taken together, these findings suggest that interventions focused on closing the achievement gap must do more than simply enroll Latino children in early education programs. Rather, there is a need to employ research-based instructional practices tailored to the needs of DLLs to promote their academic learning and school readiness (Goldenberg, 2008; National Task Force on Early Childhood Education for Hispanics, 2007; *The Future of Children*, 2005). Pre-kindergarten is considered a critical period for Latino children, many of whom face the difficult task of simultaneously learning a new language while acquiring essential school readiness skills such as getting along with others, identifying the names of letters and numbers, and differentiating letter sounds. Unfortunately, there has been little systematic attention paid to optimal early education programs for Latino DLLs and the specific classroom practices that would foster language and literacy learning in this population. The National Task Force on Early Childhood Education for Hispanics called this one of the most important unanswered questions within the early childhood field.

Within the broader education field, several research syntheses have examined the effects of specific instructional practices on academic outcomes for Latino dual language learners (see for example, August & Shanahan, 2006; Gersten et al., 2007; Goldenberg, 2008; Slavin & Cheung, 2005). These research syntheses offer three major conclusions that apply primarily to school-age children, but also point to promising practices that require further evaluation with a younger group of children enrolled in early education programs.

The first conclusion from the literature addresses the importance of supporting the child's primary language, also viewed by many early childhood experts and practitioners as a critical component of language development and school readiness for DLLs (Buysse et al., 2005; Garcia, Kleifgen, & Falchi, 2008; Gillanders & Castro, 2007; Maschinot, 2008; NAEYC, 1996; Sanchez, 1999). It is now widely recognized that learning two languages at the same time does not cause confusion or language delays in young children, and that supporting the development of both languages actually facilitates English language learning. Furthermore, mounting empirical evidence supports this conclusion (see review of the literature by Maschinot, 2008). As a result, oral and written language experiences for DLLs should be regarded as an additive process to ensure that children are able to maintain their home language while also learning to speak and read English (August & Hakuta, 1997; August & Shanahan, 2006; Bialystok, 2001; Snow, 2006).

The second conclusion addressing instructional strategies for DLLs is that an effective curriculum and sound instruction serve as the foundation of learning for every child, including DLLs. However, a study sponsored by the National Center for Education Research (2008) that conducted a rigorous evaluation of available preschool curricula (the Preschool Curriculum Evaluation Research [PCER] initiative) found that only one of the 14 curricular interventions positively affected student outcomes in language and literacy. Moreover, the intervention associated with positive language and literacy outcomes consisted of a combination of two different curricula, rather than a single curriculum. Despite limited evidence for the effectiveness of any single curriculum in early childhood, there is a growing awareness that it is necessary to provide all learners with the best available educational practices prior to determining whether some children have learning difficulties and may benefit from receiving additional instructional supports. This idea is a foundational principle of Response to Intervention (RTI; Jimmerson, Burns, & VanDerHeyden, 2007) and other tiered models in early childhood that are designed to enhance learning for all students by linking assessments with instruction and interventions (Brown, Odom, & Conroy, 2001; Hemmeter, Ostrosky, & Fox, 2006; *Recognition & Response Implementation Guide*, 2008; Sandall & Schwartz, 2008).

The third conclusion from research syntheses on effective instructional strategies for DLLs concerns the importance of differentiated instruction and accommodations for children who require additional supports to learn. These recommendations also derive from the principles of RTI and are consistent with the principles of tiered models in early education. These approaches are intended to guide instruction to ensure that children receive core, strategic, and sometimes intensive

supports to help them learn, and that decisions about what supports to provide are based on children's level and rate of progress. According to an IES Practice Guide on RTI, differentiated instruction is more effective when based on an assessment of students' current reading level and when implemented with a high level of fidelity (Gersten et al., 2009). Examples of differentiated reading instruction for school-age children might include providing extra practice with vocabulary words with a partner, instructing small groups of children to identify the characteristics and motivations of main characters in books they are reading, and allowing children to play games that emphasize phoneme segmentation or blending. Unfortunately, scant research exists across all age groups to help teachers determine precisely what interventions and which instructional accommodations and adaptations are most beneficial for Latino DLLs (Snow, 2006).

Although findings from previous research provide some general guidance and recommendations for improving instructional practices for Latino DLLs primarily in the elementary grades, there is a paucity of research on these issues for pre-kindergarten children. Early childhood programs are largely unprepared to address the diverse educational and linguistic needs of Latino DLLs and their families. The limited research on this topic suggests that early childhood professionals may lack knowledge of the cultural beliefs and values that Latino families from diverse backgrounds find important, and may lack understanding of the strengths and variations of a child-rearing system that is different from their own (Buysse et al., 2005; Fuller, Egger-Pierola, Holloway, Liang, & Rambaud, 1996; Harry, 1992). Furthermore, and more relevant for this investigation, early childhood teachers generally lack appropriate and effective methods of adapting their instructional practices to support language and literacy skills among Latino DLLs. The literature suggests that the provision of high-quality early education will require that early childhood professionals learn new knowledge and skills related to differentiated instruction to facilitate language development and literacy learning among children whose primary language is not English (Goldenberg, 2008).

The purpose of this research study was to conduct a preliminary investigation of the *Nuestros Niños* Early Language and Literacy Program. The *Nuestros Niños* Program consists of intensive, ongoing professional development for pre-kindergarten teachers who serve Latino DLLs whose primary language is Spanish. The content of the professional development was designed to complement a core classroom curriculum and includes specific instructional strategies and supports to address language and literacy skills of DLLs. A randomized, controlled study was conducted to assess the effects of the *Nuestros Niños* program on teachers' classroom practices and children's outcomes related to language proficiency and early literacy skills in both English and Spanish. As far as we know, this was one of the first experimental studies conducted to examine the efficacy of a professional development intervention targeting early language and literacy skills of Latino DLLs in pre-K. The study addressed two primary research questions. First, do teachers in pre-K classrooms receiving the *Nuestros Niños* intervention make greater improvements in their classroom practices related to facilitating language and literacy learning than teachers in control classrooms? Second, do Latino DLLs in pre-k classrooms receiving the *Nuestros Niños* intervention make greater gains in language and literacy skills than their peers in control classrooms?

1. Method

1.1. Participants and settings

The study was conducted with 55 lead teachers and 193 Latino pre-kindergarten DLLs and their families. All 55 lead teachers were monolingual English-speakers with few classrooms (35% of intervention classrooms, 21% of control classrooms) having a bilingual assistant teacher. Furthermore, English was the primary language of instruction in all classrooms in both the intervention and control conditions. The Latino ELLs who participated in the study were at various stages in acquiring English as their second language. The percentage of children for whom Spanish was reported to be the primary language spoken at home was similar across groups (65% for those in the intervention group, 64% for those in the control group). Participants were selected from the North Carolina More at Four Pre-Kindergarten Program (MAF), a state-funded pre-kindergarten program targeting at-risk four-year-olds (particularly those not previously enrolled in early education), with a focus on promoting school readiness skills. At the time of this study, MAF served 17,000–20,000 children who were eligible on the basis of a combination of risk factors that included poverty, disability, health status, and English proficiency. The MAF pre-kindergarten model encompasses various types of early childhood programs: public school-based programs, Head Start, and child care centers (both for-profit and nonprofit). Consistent with the MAF model, classrooms from all of these types of early childhood programs were included in this study. Regardless of program type, MAF classrooms are required to meet specific standards that address staff education, class size, staff-child ratios, the curriculum, and program quality based a score of at least 4.5 on the Early Childhood Environment Rating Scale (ECERS-R; Harms, Clifford, & Cryer, 1998), a widely recognized measure of program quality and the basis of North Carolina's rated license.

The following study inclusion criteria were used to control for any further variability in classroom characteristics across program type: (1) teachers with a bachelor's degree or an associate's degree and working towards a bachelor's; (2) programs with moderate to high quality as measured by the NC star-rating system (3–5 stars out of 5 stars total), (3) use of the Creative Curriculum (a state-approved MAF curriculum and the predominant curriculum used by MAF classrooms), (4) classroom enrollment of at least four Latino ELL children, but not to exceed 85% of total enrollment, and (5) use of English as the primary language of instruction.

Participating teachers were asked to distribute parent permission forms to the parents of all Latino DLLs enrolled in their class, and four children with parent permission were randomly selected to participate in each classroom. Classrooms were

Table 1
Demographic characteristics of participants and settings.

Child	Control (<i>n</i> = 101)				Intervention (<i>n</i> = 92)			
	%	<i>M</i>	<i>SD</i>	Range	%	<i>M</i>	<i>SD</i>	Range
Age in months	–	54.3	3.7	47–62	–	54.8	4.4	38–70
Sex (female)	55	–	–	–	66	–	–	–
Family's country of origin								
U.S.	4	–	–	–	7	–	–	–
Mexico	79	–	–	–	80	–	–	–
Puerto Rico	1	–	–	–	0	–	–	–
Other	16	–	–	–	13	–	–	–
Family's years of residence in U.S.	–	8.9	4.65	1.5–25	–	8.4	4.27	.1–23
% Spanish spoken to child at home	–	64.4	33.2	0–100	–	65.2	30.34	0–100
Classroom	Control (<i>n</i> = 29)				Intervention (<i>n</i> = 26)			
	%	<i>M</i>	<i>SD</i>	Range	%	<i>M</i>	<i>SD</i>	Range
Type of program								
Child care	45	–	–	–	46	–	–	–
Public pre-K	27.5	–	–	–	35	–	–	–
Head Start/blended	27.5	–	–	–	19	–	–	–
Staff–child ratio	–	1:8	–	1:9 – 1:4	–	1:8	–	1:9 – 1:5
% Latino child enrollment	–	46.6	16.9	16.6–76.5	–	41.6	16.6	22.2–87.5
Bilingual lead or assistant teacher	21	–	–	–	35	–	–	–
NAEYC accredited	31	–	–	–	27	–	–	–
Lead teacher race/ethnicity								
Asian	0	–	–	–	4	–	–	–
Black	34	–	–	–	54	–	–	–
Hispanic/Latino	7	–	–	–	0	–	–	–
White	59	–	–	–	42	–	–	–
Lead teacher education								
Associate (CDA)	10	–	–	–	11	–	–	–
Bachelors	73	–	–	–	81	–	–	–
Master	17	–	–	–	8	–	–	–

then randomly assigned to the intervention (*n* = 26 teachers; *n* = 92 children) or control (*n* = 29 teachers; *n* = 101 children) group. For cases in which there were two participating classrooms from the same program, these classrooms were assigned to the same study condition. Teachers in the intervention group participated in the professional development intervention throughout the school year. Teachers in the control group were invited to participate in the professional development institutes at the end of the study. Table 1 provides a summary of the demographic characteristics of the participating children and families, programs, and teachers. Analyses based on *t*-tests and chi-square indicated that there were no statistically significant differences between intervention and control groups related to child, teacher, and classroom variables.

1.2. Intervention

The content of the Nuestros Niños program is based on the best available research evidence on effective instructional practices to promote language and literacy skills (e.g., phonological awareness, concepts of print, oral language, letter identification, alphabetic principle) in pre-kindergarten children in general, as well as the more limited literature on this topic to address the specific learning characteristics of Latino DLLs (Espinosa, Castro, Crawford, & Gillanders, 2007). The instructional practices were designed to complement the core curriculum and to provide monolingual English-speaking teachers who use English as the primary language of instruction with specific accommodations for DLLs (e.g., the use of visual cues and props, implementing pre-reading activities prior to group storybook time, systematically observing and documenting second language learning). The format of the Nuestros Niños program reflects research-based conclusions suggesting that professional development is more likely to be effective when it is (a) practice-specific, (b) aligned with instructional goals and the curriculum that teachers use in practice, and (c) designed to offer ongoing guidance and feedback in practice settings (Buysse, Winton, & Rous, 2009; Cochran-Smith & Zeichner, 2005; National Professional Development Center on Inclusion, 2008). In accordance with these recommendations, the Nuestros Niños program consisted of three professional development components: (1) professional development institutes to promote teachers' acquisition of core content knowledge and skills, (2) individualized consultation sessions to support teachers in implementing new instructional strategies in the classroom, and (3) community of practice meetings to provide participating teachers with opportunities for feedback, reflection, and collaborative problem-solving.

Teachers participated in a three-day institute at the beginning of the school year conducted over a period of several weeks. Following the institutes, teachers were assigned to work with one of two bilingual Latina consultants (who were also responsible for facilitating the three-day institutes and community of practice meetings). Each consultant worked with a group of 6–8 teachers for eight weeks, visiting individual teachers once every other week and conducting community of practice meetings with her assigned group of teachers on alternate weeks. The consultants had graduate degrees in an early childhood-related field. They participated in intensive professional development and once-a-week reflective supervision meetings, both facilitated by the research team, to support their consultation practice and facilitation of the community of practice meetings.

1.2.1. *Nuestros Niños Professional Development Institutes*

The content of the institutes is presented in the *Nuestros Niños Handbook on Early Language and Literacy* (Castro, Gillanders, Machado-Casas, & Buysse, 2006). The handbook was validated through a review by national experts in early childhood education and second language learning. It contains the following six modules that include core concepts, instructional practices, classroom resources, and specific accommodations for promoting language and literacy development in both English and Spanish among Latino DLLs: (1) foundations for teaching language and literacy to dual language learners; (2) bilingual oral language development in young children; (3) early literacy; (4) early reading; (5) early writing; and (6) assessment of early language and literacy in young dual language learners. In addition to making books and print materials available to children in both English and Spanish, we provided teachers with a range of instructional strategies to support language and literacy development in English, as well as suggesting specific accommodations to facilitate the development of children's primary language in Spanish whenever possible. These strategies included using visual cues and props to introduce new vocabulary, the dramatization of story narratives, teaching in small groups, and obtaining assistance from either bilingual teaching assistants or classroom volunteers in implementing instructional strategies, such as book reading in both English and Spanish or incorporating the core vocabulary in both English and Spanish.

1.2.2. *Nuestros Niños consultation model*

The consultation approach was adapted from the eight-stage consultation model for early childhood programs developed and manualized by Buysse and Wesley (see for example, Buysse & Wesley, 2004, 2005; Wesley & Buysse, 2004) to help teachers develop and implement an action plan that is based on assessment and focused on implementing key instructional strategies for dual language learners. Contact summary forms describing the nature and intensity of the consultation sessions, and weekly reflective supervision meetings with members of the research team were used to ensure that consultants were implementing the consultation model with fidelity. An implementation plan developed jointly by each classroom teacher and her consultant was used as the basis of bimonthly visits to monitor a teacher's progress and make necessary adjustments related to project goals for improving language and literacy learning for Latino DLLs.

1.2.3. *Nuestros Niños Community of Practice Model*

This approach combines principles of situated learning and reflective practice with the Lesson Study approach to offer teachers a structured process for developing and refining instructional strategies through shared inquiry and learning (Buysse, Sparkman, & Wesley, 2003; Wesley & Buysse, 2006). Through regular community of practice meetings (six–eight, once every other week), teachers received guidance in working together to create a lesson around a commonly agreed upon goal related to ELL children's language and literacy learning and reflecting key concepts and strategies from the *Nuestros Niños* program. The process involves (1) identifying a topic and creating a lesson plan in collaboration with others (e.g., an interactive read aloud activity), (2) implementing the lesson in the classroom while a colleague observes or the lesson is videotaped, (3) reflecting on implementation with members of the community of practice, (4) revising the lesson on the basis of reflection and feedback, and (5) creating a product for dissemination to the broader educational community. The format of the community of practice meetings provided teachers with opportunities to view videotapes of their own practices and those of other teachers as a method of determining how they could refine and improve their instructional strategies for Latino ELL children.

1.3. *Data collection measures and procedures*

The Early Language and Literacy Classroom Observation (ELLCO) Toolkit (Education Development Center, 2002) is a standardized measure that was used to assess the quality of the classroom practices. The ELLCO contains three subscales: the classroom observation scale (14 items rated on a 1–5 scale), the literacy environment checklist (24 items scored as yes or no, or number of occurrences), and the literacy activities rating scale (nine items scored as yes or no, or the number of occurrences). The ELLCO Addendum (Castro, 2005) was developed as a companion measure to assess specific practices targeting DLLs for each of the three subscales of the ELLCO: the classroom observation scale (eight items rated on a 1–5 scale), the literacy environment checklist (10 items scored as yes or no, or the number of occurrences), and the literacy activities rating scale (seven items scored as yes or no, or the number of occurrences). Examples of ELLCO Addendum items include rating the extent to which teachers used multiple strategies to support comprehension during book reading (the classroom observation scale), documenting the number of books and print materials available in Spanish (the literacy environment checklist), and assessing the extent to which teachers used spoken Spanish during small and large group activities (the literacy activities

rating scale.) Child knowledge and skills related to language proficiency, vocabulary, letter identification, and phonological awareness were assessed in both English and Spanish with the following measures: the Woodcock Language Proficiency Battery-Revised: English and Spanish Forms (WLPB-R; Woodcock, 1991; Woodcock & Muñoz-Sandoval, 1995); the Peabody Picture Vocabulary Test (PPVT-III; Dunn & Dunn, 1997) and corresponding Test de Vocabulario en Imágenes Peabody (TVIP; Dunn, Padilla, Lugo, & Dunn, 1986); the Phonological Awareness Tasks (PAT; Miccio & Hammer, 2002); Naming Letters (National Center for Early Development & Learning, 2003); and Where's My Teddy Story and Print Concepts (FACES: The Head Start Child and Family Experiences Survey, 2003).

Bilingual data collectors who were blind to the study conditions were trained on standardized assessment and classroom observation measures by experienced members of the research team using widely recognized training methods (e.g., supervised child assessment practice sessions with feedback, refresher training prior to spring data collection). Classroom observations were conducted over a 4–6 h period in all intervention and control classrooms, once in the fall and again in the spring. Parallel child assessments in English and Spanish were administered individually to all 193 children 1–2 weeks apart in all intervention and control classrooms, once in the fall and again in the spring. The language used in child assessment was counterbalanced to control for the order of administration. Different data collectors gathered the classroom observations and child assessments within each classroom at both data collection points.

Interrater reliability data for the ELLCO and ELLCO-A were gathered throughout the data collection period on approximately 20% of the classrooms. Cohen's kappa coefficients were calculated for each item on the classroom observation scale on both measures, with overall mean values of .44 for ELLCO and .46 for ELLCO-A. Percent exact agreement was calculated for each item on the literacy environment checklist on both measures, with overall mean values of 91% on ELLCO and 94% on the ELLCO-A. Percent exact agreement was calculated for each item on the literacy activities rating scale on both measures, with overall mean values of 94% on ELLCO and 100% on the ELLCO-A.

2. Results

2.1. Analysis plan

Analyses were conducted to address the primary research questions: (1) are there greater improvements over the school year in classroom practices for the intervention than control classrooms; and (2) do children in intervention classrooms make greater gains over the school year in language and literacy skills than children in control classrooms?

To address the first research question, we conducted separate general linear models analyses for each classroom practices score on the ELLCO (classroom observation scale total, literacy activities rating scale total, literacy environment checklist total) and the ELLCO Addendum (Addendum Classroom Observation Scale total, addendum literacy activities rating scale total, addendum literacy environment checklist total). Difference scores (spring score minus fall score) were calculated for each classroom on each outcome measure, and these were used as the dependent variables. This approach was utilized, given the relatively small classroom sample size, because the difference scores exhibited a fairly normal distribution while the fall and spring scores exhibited quite skewed distributions. Treatment group (intervention vs. control) was included as the independent variable and two classroom characteristics, percentage of Latino children and teacher education level (coded as a three-level variable: Associate's, Bachelor's, or Master's degree) were included as covariates. A reference cell coding was used, with control group classrooms with AA level teachers used as the reference cell. Cohen's *d* was used to calculate effect size estimates based on the differences between intervention and control group adjusted mean difference scores.

To address the second research question about differences in children's language and literacy growth in relation to participation in the intervention, we conducted separate mixed models analyses for each child outcome in Spanish and in English, including the WM/WJ Broad Language Proficiency score, the TVIP/PPVT receptive language standard score, the PAT phonological awareness total, the Naming Letters task total score, and the Story concepts task total score. In the case of significant findings for measures with subscale areas (the WM/WJ language proficiency and the PAT phonological awareness), a second set of analyses examining the subscale scores was conducted to determine the specific areas of influence. Difference scores (spring score minus fall score) were calculated for each child and were used as the dependent variables; given the relatively small classroom sample size, the difference scores were fairly normally distributed while the distributions for fall and spring scores were quite skewed. Treatment group (intervention vs. control) was included as the independent variable and covariates included two classroom characteristics, percentage of Latino children and teacher education level (coded as a three-level variable: Associate's, Bachelor's, or Master's degree), as well as child characteristics (gender in all analyses and age in analyses not using age-standardized scores). These analyses also adjusted for classroom, to control for dependencies in the data from children in the same classroom. A reference cell coding was used, with males in control group classrooms with AA level teachers used as the reference cell. Cohen's *d* was used to calculate effect size estimates based on the differences between intervention and control group adjusted mean difference scores.

2.2. Results for classroom practices

Table 2 shows the mean scores on the fall and spring classroom practices measures for intervention and control group classrooms. Changes in classroom language and literacy practices were examined as a function of participation in the inter-

Table 2

Means, standard deviations, and effect sizes for classroom practices measures.

Measure	Intervention (<i>n</i> = 26)		Control (<i>n</i> = 29)		Effect size ^a
	Fall, <i>M</i> (SD)	Spring, <i>M</i> (SD)	Fall, <i>M</i> (SD)	Spring, <i>M</i> (SD)	
ELLCO					
Classroom Observation Scale ^b	3.2 (.4)	3.3 (.6)	3.3 (.6)	3.3 (.5)	.29
Literacy Environment Checklist ^c	25.6 (4.8)	28.2 (5.1)	26.0 (5.8)	28.6 (6.40)	.00
Literacy Activities Rating Scale ^d	6.2 (2.9)	7.0 (3.0)	7.3 (3.1)	6.1 (2.9)	.63*
ELLCO Addendum					
Classroom Observation Scale ^b	2.2 (.5)	2.5 (.6)	2.4 (.5)	2.4 (.4)	.61*
Literacy Environment Checklist ^e	2.8 (1.5)	5.0 (2.4)	2.7 (1.6)	3.4 (1.8)	.81**
Literacy Activities Rating Scale ^f	.8 (1.3)	.5 (.9)	.6 (1.1)	.4 (.9)	-.16

^a Effect size estimates (Cohen's *d*) were computed as the difference between the intervention and control groups in the least-square mean difference scores (gains/losses from fall to spring) adjusted for percentage of Latino children and teacher education level based on general linear models analyses.

^b Represents mean item scores on a 1–5 scale.

^c Total possible range = 0–41.

^d Total possible range = 0–13.

^e Total possible range = 0–16.

^f Total possible range = 0–14.

* *p* < .05.

** *p* < .01

vention versus control group, adjusting for teacher education levels and the percentage of Latino children. Table 2 displays the effect size estimates for these analyses and Table 3 provides the parameter estimates. Intervention classrooms had greater gains than control classrooms on three of the six scales: ELLCO literacy activities rating scale, addendum classroom observation scale, and addendum literacy environment checklist. The effect sizes for all three of these scales were in the moderate to large range. Examination of the adjusted mean difference scores for the ELLCO literacy activities rating scale indicate that intervention classrooms increased the general frequency of literacy activities from fall to spring (*M* = .2), while control group classrooms showed decreases in the frequency of such activities over this same time period (*M* = –1.8). In terms of the Addendum ratings, these findings indicate that while both groups showed improvement over the school year, the intervention classrooms made significantly greater improvements as compared to the control classrooms in both the quality of instructional practices (intervention *M* = 3.5; control *M* = .9) and the quality of the literacy environment (intervention *M* = 2.6; control *M* = .9).

Table 3

Regression analyses summary for the classroom practices measures.

Parameter	ELLCO Classroom Observation Scale			ELLCO Literacy Environment Checklist			ELLCO Literacy Activities Rating Scale		
	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>
Intercept	.35	.21	.10	2.72	1.84	.15	–.55	1.23	.66
Study condition									
Control	.00	–	–	.00	–	–	.00	–	–
Intervention	.16	.15	.30	.01	1.31	.99	1.99	.88	.03
% Latino	–.00	.00	.64	–.01	.04	.74	–.03	.03	.32
Teacher education									
BA	–.44	.22	.06	–.12	1.96	.95	–.42	1.31	.75
MA	–.20	.30	.51	–.85	2.69	.75	–3.19	1.80	.08
AA	.00	–	–	.00	–	–	.00	–	–
Parameter	ELLCO Addendum Classroom Observation Scale			ELLCO Addendum Literacy Environment Checklist			ELLCO Addendum Literacy Activities Rating Scale		
	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>
Intercept	.22	.22	.32	1.86	.79	.18	–.20	.58	.73
Study condition									
Control	.00	–	–	.00	–	–	.00	–	–
Intervention	.32	.15	.04	1.66	.57	.01	–2.44	.42	.56
% Latino	.00	.00	.54	.01	.02	.42	–.02	.01	.21
Teacher education									
BA	–.25	.23	.29	–.55	.85	.52	.12	.61	.85
MA	–.06	.31	.84	.05	1.16	.97	.18	.84	.83
AA	.00	–	–	.00	–	–	.00	–	–

Table 4

Means, standard deviations, and effect sizes for child outcome measures.

Measure	Intervention (<i>n</i> = 92)		Control (<i>n</i> = 101)		Effect size ^a
	Fall, <i>M</i> (<i>SD</i>)	Spring, <i>M</i> (<i>SD</i>)	Fall, <i>M</i> (<i>SD</i>)	Spring, <i>M</i> (<i>SD</i>)	
Spanish child outcomes					
WLPB Broad Language	75.8 (10.4)	74.0 (11.0)	77.0 (10.8)	73.4 (10.5)	.41
TVIP Receptive Language	84.0 (15.7)	83.0 (19.6)	81.3 (14.4)	81.2 (16.7)	.10
Phonemic Awareness Task (PAT)	13.2 (5.3)	16.2 (6.3)	13.9 (4.6)	14.8 (4.9)	.69 [*]
PAT Rhyme Matching	4.7 (2.8)	6.0 (2.3)	4.9 (2.4)	5.2 (2.4)	.68 [*]
PAT Onset Matching	2.7 (1.9)	3.5 (2.2)	3.0 (1.9)	3.1 (2.0)	.41
PAT Onset Segmentation and Matching	2.7 (1.7)	3.4 (2.1)	3.0 (1.8)	3.4 (1.9)	.03
Naming Letters Task	.6 (2.5)	1.6 (3.7)	.5 (1.7)	1.0 (3.2)	.48
Story Concepts Task	3.3 (2.2)	4.3 (2.3)	3.0 (2.1)	4.4 (2.3)	−.34
English child outcomes					
WLPB Broad Language	69.5 (15.1)	74.2 (16.0)	69.2 (12.6)	74.9 (15.0)	−.15
PPVT-III Receptive Language	60.6 (17.3)	65.0 (18.0)	62.6 (17.6)	68.1 (18.4)	−.11
Phonemic Awareness Task (PAT) ^b	9.5 (3.9)	11.7 (5.3)	9.5 (4.5)	10.5 (4.7)	.39
PAT Rhyme Matching	4.8 (2.7)	5.6 (3.0)	4.1 (2.9)	5.3 (2.5)	−.20
PAT Onset Matching	2.7 (1.9)	3.4 (2.2)	2.8 (1.9)	2.9 (2.1)	.41
PAT Onset Segmentation and Matching	2.0 (1.5)	2.8 (2.1)	2.4 (1.9)	2.3 (1.7)	.61 [*]
Naming Letters Task	3.1 (6.0)	9.9 (8.6)	2.8 (5.1)	9.5 (8.6)	.11
Story Concepts Task	2.5 (2.2)	3.9 (2.6)	2.4 (2.2)	3.6 (2.4)	.23

^a Effect size estimates (Cohen's *d*) were computed as the difference between the intervention and control groups in the least-square mean difference scores (gains/losses from fall to spring) adjusted for classroom, percentage of Latino children, teacher education level, child gender, and child age for non-age standardized scores, based on mixed models analyses.

^b Cronbach alpha coefficients in English were .52 overall, .42 for rhyme matching, .35 for onset matching, and .19 for onset segmentation and matching. Cronbach alpha coefficients in Spanish were .59 overall, .42 for rhyme matching, .30 for onset matching, and .22 for onset segmentation and matching.

* *p* < .05.

2.3. Child outcomes

Table 4 shows the mean scores on the fall and spring child outcome measures for intervention and control group classrooms. Differences in children's gains in language and literacy skills over the school year were investigated as a function of participation in intervention versus control classrooms, after adjusting for classroom characteristics (teacher education and percentage of Latino children) and child characteristics (gender and age for non-standardized measures). Table 4 provides the effect size estimates for these analyses, along with alpha coefficients for the Phonological Awareness Task scales which were .52 overall in English (.19–.42 for subscales) and .59 overall in Spanish (.22–.42 for subscales). Table 5 displays the parameter estimates. For outcomes assessed in Spanish, there were greater gains for children in intervention than control classrooms on one of the measures, the Phonological Awareness Task, but not on the other four measures. Examination of the adjusted mean difference scores shows that children made gains in both intervention and control group classrooms, but the gains were substantially larger in the intervention classrooms (*M* = 2.3) than in control classrooms (*M* = .4). Given this significant finding, an additional set of analyses was conducted on the three phonological awareness subscales (rhyme matching, onset matching, and onset segmentation and matching) to determine the skills most affected. These analyses indicated that the significant effects were primarily due to greater gains in rhyming skills (rhyme matching) for children in the intervention classrooms. Again, children made gains from fall to spring in both groups, but the gains based on adjusted mean scores were larger for children in intervention (*M* = 1.2) versus control classrooms (*M* = .1). The effect sizes for both the overall PAT score and the rhyme matching subscale score were in the moderate to large range.

For outcomes assessed in English, there were no significant differences between intervention and control group classrooms in children's growth on the overall scores for the language and literacy measures. Given the significant effects on the Phonological Awareness Task in Spanish, we conducted a parallel set of additional analyses of the subscales in English. These analyses revealed a significant effect in the moderate to large range for one subscale, PAT onset segmentation and matching, indicating that children in intervention classrooms made greater gains than children in control classrooms. Examination of the adjusted mean difference scores indicated that children in intervention classrooms exhibited growth from fall to spring in these skills (*M* = .8), while children in control classrooms exhibited almost no change (*M* = -.0).

3. Discussion

This study consisted of a preliminary investigation of the effects of the Nuestros Niños professional development program on classroom practices and child outcomes for Latino DLLs. As such, it represents an initial attempt to answer a critically important question within the early childhood field, namely, whether specific instructional practices acquired through professional development are effective for promoting language and literacy learning among this population of young children. Although additional research is needed to address this question more fully, this study produced several findings that

Table 5

Regression analysis summary for child outcome measures.

Parameter	English measures											
	WLPB Broad Language			PPVT Receptive Language			Naming Letters Task			Story Concepts Task		
	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>
Intercept	4.01	2.96	.18	1.52	3.09	.62	4.62	1.84	.02	1.30	.55	.02
Gender												
Female	−.38	1.56	.81	1.03	1.78	.57	.88	1.01	.39	.24	.31	.45
Male	.00	–	–	.00	–	–	.00	–	–	.00	–	–
Age	–	–	–	–	–	–	−.03	.12	.83	−.04	.04	.26
Study condition												
Intervention	−.90	1.70	.60	−.70	1.78	.70	.43	1.10	.70	.27	.33	.42
Control	.00	–	–	.00	–	–	.00	–	–	.00	–	–
% Latino	.04	.05	.48	.03	.05	.60	.04	.03	.25	.01	.01	.56
Teacher education												
BA	1.05	2.93	.72	2.44	2.96	.41	2.18	1.78	.23	.01	.54	.99
MA	1.41	3.53	.69	4.05	3.65	.27	1.33	2.23	.55	.14	.67	.84
AA	.00	–	–	.00	–	–	.00	–	–	.00	–	–
Parameter	PAT Total			PAT Rhyme Matching			PAT Onset Matching			PAT Onset Segmentation Matching		
	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>
	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>
Intercept	1.88	1.28	.15	.79	.95	.41	.57	.64	.38	.86	.60	.16
Gender												
Female	−.71	.75	.35	−.26	.49	.60	−.32	.36	.38	−.26	.35	.47
Male	.00	–	–	.00	–	–	.00	–	–	.00	–	–
Age	−.06	.09	.49	−.03	.06	.62	.02	.04	.59	−.06	.04	.12
Study condition												
Intervention	1.06	.77	.17	−.41	.57	.48	.55	.38	.16	.78	.36	.04
Control	.00	–	–	.00	–	–	.00	–	–	.00	–	–
% Latino	−.01	.02	.58	.00	.02	.89	.00	.01	.79	−.02	.01	.16
Teacher education												
BA	.94	1.23	.45	.25	.92	.79	.43	.62	.49	−.01	.58	.98
MA	−.40	1.57	.80	−.03	1.16	.98	−.10	.79	.90	.04	.73	.95
AA	.00	–	–	.00	–	–	.00	–	–	.00	–	–
Parameter	Spanish measures											
	WLPB Broad Language			TVIP Receptive Language			Naming Letters Task			Story Concepts Task		
	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>
Intercept	−1.12	2.82	.69	3.89	3.31	.25	.93	.53	.09	.22	.50	.67
Gender												
Female	−.96	1.57	.55	−1.33	2.02	.52	−.35	.34	.31	−.13	.29	.66
Male	.00	–	–	.00	–	–	.00	–	–	.00	–	–
Age	–	–	–	–	–	–	.02	.04	.67	.00	.03	.92
Study condition												
Intervention	2.47	1.71	.16	.70	2.06	.74	.55	.33	.10	−.36	.31	.24
Control	.00	–	–	.00	–	–	.00	–	–	.00	–	–
% Latino	.06	.05	.24	.18	.06	.01	.02	.01	.12	.01	.01	.56
Teacher education												
BA	−.23	2.75	.93	−4.00	3.22	.22	.28	.52	.59	1.03	.49	.04
MA	3.10	3.53	.38	−2.53	4.21	.55	.36	.66	.59	1.12	.62	.08
AA	.00	–	–	.00	–	–	.00	–	–	.00	–	–
Parameter	PAT Total			PAT Rhyme Matching			PAT Onset Matching			PAT Onset Segmentation Matching		
	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>
	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>	β	SE	<i>p</i>
Intercept	4.21	1.25	.00	1.64	.75	.03	.80	.51	.12	.14	.58	.80
Gender												
Female	−1.43	.80	.08	−.62	.44	.18	.32	.33	.34	−.49	.36	.19
Male	.00	–	–	.00	–	–	.00	–	–	.00	–	–
Age	.11	.10	.27	−.03	.05	.53	.06	.04	.10	−.01	.04	.75

Table 5 (Continued)

Study condition												
Intervention	1.91	.78	.02	1.11	.46	.02	.44	.31	.16	.04	.36	.91
Control	.00	–	–	.00	–	–	.00	–	–	.00	–	–
% Latino	.01	.02	.76	.02	.01	.17	–.01	.01	.18	–.02	.01	.16
Teacher education												
BA	–.50	1.21	.68	.04	.73	.96	–.25	.50	.62	.85	.57	.14
MA	–2.90	1.57	.07	–.45	.95	.64	–.63	.63	.32	–.40	.72	.58
AA	.00	–	–	.00	–	–	.00	–	–	.00	–	–

can inform future research designed to explicate issues around how teachers' attempts to provide targeted, differentiated instruction intersect with Latino children's second language learning.

Study findings were examined and can be interpreted at two levels: classroom practices and child outcomes. With regard to classroom practices, the primary research question was whether the professional development produced greater gains over a school year in the quality of classroom practices for the intervention versus control classrooms. The results showed that intervention classrooms made greater gains than control classrooms on three of the six quality practice scales (ELLCO literacy activities rating scale, ELLCO Addendum classroom observation scale, and ELLCO Addendum literacy environment checklist), and that the difference between intervention and control classrooms was statistically significant. The effect sizes for these results were in the moderate to large range. In addition to finding that intervention classrooms made greater gains in increasing classroom-wide literacy activities from fall to spring, the results indicated that control classrooms actually showed a decrease in classroom-wide literacy activities over the same period. Moreover, the findings related to the Addendum scales showed that intervention classroom teachers made greater improvements than control classroom teachers in setting up a supportive environment and implementing activities that were specifically designed to address the language and literacy needs of Latino DLLs (e.g., incorporating Spanish books and other Spanish print materials into interest centers).

With respect to child outcomes, the primary research question was whether children in the intervention classrooms would make greater gains over the school year in language and literacy skills than children in the control classrooms. The study employed a parallel assessment method to assess child outcomes in both English and Spanish. For outcomes assessed in Spanish, there were greater gains that were statistically significant for children in intervention versus control classrooms on one of the five measures, the Phonological Awareness Task. Follow-up analyses revealed that the positive effect on phonological awareness in Spanish was due primarily to greater gains in the area of rhyming skills. Effect sizes for both the overall phonological awareness measure and the rhyming subscale were in the moderate to large range. However, the low internal consistency of this measure across the three scales represents a limitation of the study.

For outcomes assessed in English, there were no statistically significant differences in children's growth on the overall scores for the language and literacy measures in intervention and control classrooms. However, a parallel analysis of the subscales of the Phonological Awareness Task in English revealed a significant effect for one subscale, onset segmentation and matching. As compared to children in control classrooms, children in the intervention classrooms made greater gains in segmenting and matching syllables in English, with the size of the effect shown to be moderate. The low internal consistency of this measure also represents a study limitation.

Taken together, these findings suggest that initial attempts to design a professional development intervention focused on instructional strategies for Latino DLLs was more effective for improving teaching practices related to language and literacy than for enhancing children's outcomes in these areas. On a number of child outcome measures, the study found no effects for the intervention. One explanation is that existing measures of language and literacy may not be sensitive enough to detect relatively short-term gains in young children in these content areas, a problem that may help to explain similar findings even within large-scale studies such as the PCER initiative. Another explanation may be that additional time was needed before changes in teaching practices could be expected to improve children's learning outcomes. It is now widely acknowledged that the effectiveness of professional development relies on a chain of causal evidence with several critical links: evidence that teachers acquire new knowledge and skills; evidence that teachers apply knowledge and skills as these were intended to be implemented in practice; and evidence that teacher practices acquired through professional development produce positive outcomes for children's learning and development (Cochran-Smith & Zeichner, 2005). Within early childhood and the broader field of education, there is still much that is unknown about each one of these causal linkages in professional development (e.g., precisely how to measure teacher knowledge and skills, how to determine and control for intervening variables at both the teacher and child levels), much less the combination of these factors. The body of research that directly addresses student outcomes as a function of professional development, and the conditions and contexts within which these outcomes occur, remains relatively small and inconclusive, according to a report of the AERA Panel on Research and Teacher Education (Cochran-Smith & Zeichner, 2005). One question that should be addressed in future research related to professional development is how much lag time it is reasonable to expect between the implementation of new classroom practices and evidence that these practices result in improved child outcomes. Teachers in this study who participated in the professional development institutes acquired new knowledge and skills, but their application of this knowledge in their classrooms continued throughout the year and was supported through consultation and community of practice meetings. Although study findings showed that teachers were able to implement new approaches that led to measurable improvements in both the overall quality of their language and literacy practices and those specific to working with Latino DLLs, it may

have been too early in the process to expect children enrolled in these classrooms to show developmental gains as a result of these improvements in classroom practices.

Although the professional development intervention resulted in relatively modest gains for children (with positive effects only on the phonological awareness measure), the content of the professional development—effective instructional strategies for promoting language and literacy in general, along with specific strategies to scaffold learning for DLLs—is consistent with recommendations in the literature on this topic. Some of these recommendations include providing explicit vocabulary instruction, monitoring children's English language development and learning, and using small-group activities to introduce or reinforce new concepts and skills (Castro, Espinosa, & Paez, *in press*; Castro, Peisner-Feinberg, Buysse, & Gillanders, *in press*). This study incorporated these as well as other methods for use by monolingual English-speaking teachers to support language and literacy development for DLLs in English, as well as provided specific accommodations to facilitate the development of children's primary language whenever possible (e.g., use of visual cues and props, introducing vocabulary in Spanish as well as English, repetition of key literacy activities, dramatization of story narratives).

In addition to these scaffolding strategies, Geva (2006) observed that the central processing framework needs to be considered and incorporated in the development of instructional strategies to support DLLs. The central processing framework describes the transferability of processing skills that are crucial for learning to read and write. Experts have observed that some aspects of learning to read, notably phonological processing, transfer easily across languages, whereas others, such as vocabulary development and decoding, are more language-specific (Bialystok, 2007). In this study, the theory of transferability in literacy learning may help to explain why children in intervention classrooms showed greater gains in acquiring phonological awareness skills in Spanish as opposed to English, even though English was the language of instruction in these settings. It may have been necessary for Latino children in the study to first process what they were learning about rhyming, alliteration, and letter sounds in their primary language before they could apply these skills in a new language. A recent study reported obtaining similar findings in a sample of young Latino DLLs related to the expression of skills in Spanish prior to English (Páez, Tabors, & López, 2007). Additional research is needed to explicate these early reading processes in Latino DLLs enrolled in pre-K, but this study raises an interesting possibility that phonological skills taught in English are processed in Spanish prior to emerging as observable skills in English.

Geva (2006) cautioned that early educators should not mistake delays in developing language and literacy skills among DLLs as necessarily indicative of a specific learning difficulty, while at the same time, recognize the need to monitor each child's progress to identify persistent language and literacy problems that do not respond to high-quality instruction and attempts to scaffold learning. Future research is needed to determine whether it may be possible to differentiate DLLs in pre-K who might benefit from additional instructional supports in language and literacy from those who may be at-risk for a specific reading disability that could require more intensive interventions. Response to Intervention and models of RTI adapted for pre-k such as Recognition & Response (R&R) hold particular promise in this regard (Coleman, Buysse, & Neitzel, 2006; Klingner, Sorrells, and Barrera, 2007; Recognition & Response Implementation Guide, 2008; Vanderwood & Nam, 2007; Vaughn, Mathes, Linan-Thompson, & Francis, 2005).

The limited research on this topic for pre-kindergarten children also suggests that there are multiple factors that can influence the effectiveness of a particular curriculum or instructional approach. Factors such as a child's exposure to his or her primary language and to English, opportunities to experience early literacy activities in multiple languages, and a variety of family characteristics (e.g., immigration status, family resources and structure, access to print materials) likely moderate the effects of early education experiences designed to promote language and literacy learning (Castro, Espinosa et al., *in press*; Castro, Peisner-Feinberg et al., *in press*). These factors should be addressed in future research on instructional strategies to enhance the learning and development of DLLs in early education.

Future research on this topic should address many other questions as well. Although there is ample empirical evidence on older ELL children to suggest that teaching reading in the primary language promotes higher levels of reading achievement in English, many questions remain about whether this approach is more beneficial for some students than others (e.g., those with higher or lower levels of English proficiency), and in what settings and under what conditions this approach should be used (Goldenberg, 2008). For children in pre-kindergarten settings, similar research questions should be examined regarding the most effective ways of supporting children's development and learning in their primary language, with a particular focus on early education programs in which English is the language of instruction. In this regard, it will be especially important to determine how characteristics of the home and early education program coincide to support or inhibit early learning. Within the early childhood field, as a result of federal initiatives such as Early Reading First and an emphasis on school readiness skills, there are now a number of curricula and instructional approaches that can be used to support children's literacy skills related to phonological awareness (e.g., rhyming, alliteration, segmenting, blending, letter-sound relationship; see for example What Works Clearinghouse available at <http://ies.ed.gov/ncee/wwc/>). However, there is a pressing need in future research to focus on curricular and instructional approaches to support oral language development for DLLs, along with additional research on the benefits of early bilingualism on cognitive development and executive function (Yoshida, 2008). Previous studies have focused on methods to promote some aspects of language development, most notably, vocabulary development using dialogic and interactive storybook reading activities (see for example, Whitehurst, 2004). However, additional research should focus on the most effective ways to support other aspects of language development (e.g., syntax, oral language proficiency) in both the primary and secondary language. A variety of approaches suggested in the literature such as first language education, bilingual education, and two-way bilingual immersion programs (designed for both dual language learners and native English speakers) should be evaluated through research to determine their effects on

children's oral language development and academic learning, along with their feasibility and acceptability in early education programs.

Although this study has some implications for designing future research and professional development on language and literacy learning for DLLs, it was subject to several limitations. The intervention was based on the best available research on effective professional development and instructional strategies for Latino DLLs; however, this investigation constituted a Goal 2 development study. As a result, the intervention was still being developed and refined while the evaluation to establish its efficacy was getting underway. This study represented an initial step in the sequence of establishing a line of research in a particular area, beginning with the development, manualization, and evaluation of an intervention under carefully controlled conditions with a relatively small sample. The goal of this initial study was limited to understanding the intervention's feasibility, acceptability, and initial efficacy. The next step should involve implementing the intervention to determine its effectiveness for a larger sample of young DLLs and teachers in other types of early education programs. Additional research also should focus on ways of strengthening Latino family involvement in early education programs and determine how family characteristics and home practices contribute to children's language and literacy learning.

The early childhood field is at an early stage of understanding the most effective ways of supporting the development and learning of DLLs through early education programs. This study was a preliminary investigation of the effects of a professional development program on classroom practices and language and literacy outcomes for Latino DLLs enrolled in early education programs. The study found that the intervention was effective for improving both general teaching practices related to language and literacy as well as those specific to working with Latino DLLs. The finding that children in the intervention condition acquired greater skills in phonological awareness in their primary language is intriguing because it is consistent with theories about the transferability of processing skills that are required in learning to read and write. Future research is needed to build the body of knowledge that ultimately will guide educators, administrators, and policy makers in designing high-quality early education experiences that address the needs of every young learner, including the growing number of Latino DLLs enrolled in early education programs throughout the U.S.

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